

Report R1 — Tier 1a exact engine: construction and validation

Krylov sector analysis of quantum cellular automata

QCA Sector Analysis project (Tier 1)

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Abstract

This report documents the exact Tier-1a pipeline that classifies quantum elementary cellular automata (QCA) by the fragmentation structure of their one-cycle evolution channel, and the validation gate it has passed. The engine computes the directed one-cycle transition graph on the 2^N computational basis states using *exact* $\mathbb{Z}[1/\sqrt{2}]$ amplitude arithmetic (no floating point, no tolerances), extracts strongly connected components (Krylov sectors for unitary rules), and records sector sizes and basin structure. Three independent checks all pass: (i) the one-step support $\text{succ}(x)$ agrees with a dense brick-wall unitary oracle on 3840 checks; (ii) the eight ground-truth regression items of the specification are reproduced exactly, including rule 150 open-boundary sector sizes up to $N = 17$; (iii) 35/35 sector-size cross-checks against the independent Julia HSF code match exactly. We also show that, for the one-cycle transition graph, the exact arithmetic is provably *redundant* — a single brick-wall cycle cannot produce interference, so naive support propagation gives the identical graph on all 258,048 tested images — yet is kept because it certifies the branch decomposition and is load-bearing downstream (Tier 1b/1d). Finally we report a physically instructive finding on the spin-flip (inversion) symmetry of rules 201 and 108.

1 Model and conventions

N qubits sit on a line, sites $i = 0, \dots, N - 1$, site 0 the least significant bit. A classical ECA rule $R \in \{0, \dots, 255\}$ is recoded into a local-channel tuple $(r_{00}, r_{01}, r_{10}, r_{11})$ with each $r_{mn} \in \{\text{I}, \text{V}, \text{D}, \text{E}\}$ indexed by the neighbour pair $(m, n) = (x_{i-1}, x_{i+1})$:

$$(c_p, c_q) = (0, 1) \rightarrow \text{I}, \quad (1, 0) \rightarrow \text{V}, \quad (0, 0) \rightarrow \text{D}, \quad (1, 1) \rightarrow \text{E},$$

where $c_p = c_{4m+n}$, $c_q = c_{4m+n+2}$ are the rule's outputs for centre 0/1. The unitary gate is $\text{V} = H$ (Hadamard); D, E are the reset (decay / excitation) channels used only for dissipative rules (Tier 1b onward). One QCA cycle is the brick-wall product: apply the local update at all *even* sites, then at all *odd* sites, controls reading the current (mid-cycle) bitmask. Two boundary conventions are implemented and always recorded: periodic (**pb**c) and open with fixed vacuum padding $x_{-1} = x_N = 0$ (**ob**c0).

A rule is *unitary* iff every $r_{mn} \in \{\text{I}, \text{V}\}$; there are exactly $2^4 = 16$ such rules, the subject of this report.

1.1 Exact amplitude arithmetic

Every amplitude produced by conditional Hadamards lies in $\mathbb{Z}[1/\sqrt{2}]$. We store a branch as a dictionary $\{x : (a, b)\}$ with one shared dyadic exponent m , amplitude $(a + b\sqrt{2})/2^m$, a, b arbitrary-precision integers. A conditional- H layer sets $m \rightarrow m + 1$ and updates non-fired entries $(a, b) \rightarrow (2a, 2b)$, fired entries $(a, b) \rightarrow \pm(2b, a)$; addition is componentwise; the exact-zero test is $a = b = 0$ (correct because $\sqrt{2}$ is irrational). No tolerance is used anywhere in the graph stage. Unitary branch norms evaluate to exactly 1 (as **Fractions**).

2 Directed graph and sectors

Nodes are bitmasks; edges are implicit via the memo-free $\text{succ}(x) = \{y : \langle y | \Phi_C(|x\rangle\langle x|) | y \rangle \neq 0\}$, the union of Kraus-branch supports after one cycle. Sectors are the strongly connected components (SCCs) of this *directed* graph. For unitary rules the channel is doubly stochastic, so every state is recurrent and each weakly connected component is a single SCC; we therefore extract sectors with a memory-light union-find over streamed edges (one `succ` pass, $O(2^N)$ integers). This reduced the peak memory of rule 150 at $N = 17$ from 2073 MB (stack-based Tarjan) to 45 MB at equal runtime, and matches the directed Tarjan result on all 16 unitary rules. An ergodic early-exit flags a rule "effectively ergodic at this N " when one sector exceeds a fraction f_{erg} of the space.

3 Is the exact arithmetic necessary? A one-cycle no-interference theorem

The engine tracks exact $\mathbb{Z}[1/\sqrt{2}]$ amplitudes and prunes any that cancel to *exactly* zero. A cheaper engine would just propagate the reachable *support*: treat each V as branching a site to both bit values, $\{x\} \rightarrow \{x \wedge 2^i, x \vee 2^i\}$, take unions, and never look at a sign. Since the naive support always contains the exact support, the only thing the amplitudes can buy is the *deletion* of an edge $x \rightarrow y$ whose amplitude $\langle y | \Phi_C | x \rangle$ vanishes by destructive interference. For the 16 unitary rules — where the sector partition is the entire deliverable — does this ever happen?

Empirically, never. Comparing $\text{succ}(x)$ against the naive support propagation over every basis state, both boundary conventions and $N = 3, \dots, 8$: for the 16 unitary rules the two agree on all 16,128 one-cycle images and the union-find sector partitions are identical; extending the sweep to all 256 rules (per Kraus branch) gives 0 mismatches on 258,048 images. No edge is ever interference-cancelled within one cycle.

Why — a depth-1 argument. A cycle is two half-layers, the even sublattice then the odd. Within a half-layer every gate acts on a distinct qubit of the sublattice being updated, and each gate's control reads only the *complementary* sublattice, which that half-layer does not touch. Conditioned on the (definite) control background, the half-layer is thus a tensor product of single-qubit gates on disjoint qubits — a depth-1 circuit — and a product of single-qubit gates never makes two computational paths reconverge: each output configuration has a unique preimage. Concretely, cancellation on a basis state y requires two source states differing in exactly the bit a V just acted on to be simultaneously present with equal or opposite amplitude; but before site i 's gate fires that bit is still definite (the input value in the even layer, the post-even value in the odd layer), so the partner is absent, and once site i branches no later gate in the layer touches bit i . Hence $\text{succ}(x)$ equals the naive support *exactly*, for every rule and every N : for building the one-cycle transition graph, the $\mathbb{Z}[1/\sqrt{2}]$ ring is provably redundant.

The guard is not vacuous. It is the once-per-site brick-wall structure, not the arithmetic, that removes interference. Touch a qubit twice in one step and it returns: $H \cdot H = \mathbb{I}$, so applying V to a fresh $|0\rangle$ and then V again yields exact support $\{0\}$ — the $|1\rangle$ amplitude $(0, 1) + (0, -1)$ cancels to $(0, 0)$ and is pruned — whereas the naive rule would report $\{0, 1\}$. A deeper local circuit, a two-site gate, or a non-brick update order would all reintroduce cancellation.

Why we keep it anyway. Three reasons, none of them the transition graph. (i) *Certification*. The exact amplitudes let the branch decomposition check itself: unitary branch norms evaluate

to exactly 1 and dissipative Kraus probabilities sum to exactly 1 (as `Fractions`), so a bug in the branching is caught immediately — a guarantee floating point cannot give. (ii) *Downstream exactness*. Tier 1b/1d build the restricted superoperator and the pair graph from these same branches, and there the amplitude *values* (not merely their support) are load-bearing: the Césaro rank and the peripheral spectrum are exact functions of the $\langle y|K_b|x\rangle$. (iii) *Architecture-agnostic correctness*. The no-interference property is a theorem about depth-1 brick-wall layers, established *post hoc*; the engine does not assume it. Keeping the ring makes the pipeline correct by construction rather than correct by a lucky architecture, and it would flag interference the moment the model changed.

4 What a sector is: reachable support, not long-time dynamics

It is important to state exactly what the sector partition does and does not certify, because it is easy to over-read. For a unitary rule the Floquet operator U (the brick-wall circuit) is *block-diagonal in the computational basis*, $U = \bigoplus_i U_i$, and the transition graph computes precisely this decomposition: each sector S_i spans a U -invariant coordinate subspace $\mathcal{H}_i = \text{span}(S_i)$ of dimension $|S_i|$, and both U and $U^\dagger$ preserve it ($x \in S_i$, $\langle y|U|x\rangle \neq 0 \Rightarrow y \in S_i$, and symmetrically), so $U_i := U|_{\mathcal{H}_i}$ is a genuine $|S_i|$ -dimensional unitary. The union-find over succ is exactly the coarsest such basis-aligned block-diagonalization — Hilbert-space fragmentation in the literal sense: the $\mathcal{H}_i$ are the minimal invariant subspaces spanned by computational-basis states, i.e. the maximal set of configurations *kinematically reachable* from a seed.

This is a statement about the *allowed* space, and nothing about the dynamics inside it. Three things the graph provably does not resolve:

- **The Krylov dimension of a seed.** $\dim \mathcal{K}(x) = \dim \text{span}\{U^k|x\rangle\}_{k \geq 0} \leq |S_i|$, with equality iff the eigenvalues of U_i that $|x\rangle$ overlaps are non-degenerate. The lattice symmetries (translation, reflection) generically force degeneracies, so $|S_i|$ *upper-bounds* — need not equal — the orbit dimension of a given basis state.
- **Ergodic vs. localized, scars, dark states.** Whether U_i thermalizes or localizes is a spectral property (level-spacing statistics, eigenvector structure) invisible to a support graph. A dark state that is a single basis vector is seen — it is a size-1 sector — but a dark *superposition* confined within a larger sector is not.
- **The long-time distribution**, which is exactly where the monitored and unmonitored experiments (§3; R7) split even for unitary rules. The *monitored* chain $M_i = |U_i|^2$ is doubly stochastic and irreducible on S_i , so its stationary distribution is *uniform* on S_i (Cesàro-averaged for periodic sectors): for the measured dynamics the sector is the whole story, kinematics equal thermodynamics. The *unmonitored* evolution equilibrates instead to the diagonal ensemble in the eigenbasis of U_i , $\bar{\rho} = \sum_\alpha |\langle \alpha|x\rangle|^2 |\alpha\rangle\langle\alpha|$, whose basis populations are uniform only if U_i thermalizes; localization, scars, or dark states make them non-uniform.

In one line: monitoring promotes the graph’s reachable set to the actual long-time support (uniform on S_i); without monitoring the graph gives only the container, and whether it is filled ergodically or trapped is a spectral question — level statistics — which the fragmentation graph is silent on and which we do not address here.

5 Validation

5.1 Dense brick-wall oracle

For every unitary rule, $N = 3, \dots, 6$, both boundary conventions, and every basis state, the support of the corresponding column of the exactly-built dense unitary equals $\text{succ}(x)$: **3840/3840**

checks pass (oracle unitarity error $< 10^{-15}$).

5.2 Ground-truth regression table (specification §8)

All eight items pass (`pytest`, 68 tests green):

Rule	Claim	Result
204 (IIII)	2^N singleton SCCs	pass, both bc
51 (VVVV)	single SCC of size 2^N	pass, both bc
150, obc0	domain-wall sectors $\binom{N+1}{w}$, w even	pass, $N = 8\text{--}17$
54	one frozen $ 0\rangle$ + one giant SCC of size $2^N - 1$	pass, $N \leq 12$
156, pbc	# sectors = $L_N + 1$ (Lucas)	pass, $N = 3\text{--}11$
22, pbc even N	3 recurrent classes, each size 1	pass
0,22,28,29	dissipative branch norms = 1 exactly	pass

For rule 150 at $N = 17$, obc0, the exact sector sizes are

$$[43758, 43758, 18564, 18564, 3060, 3060, 153, 153, 1, 1] = \left\{ \binom{18}{w} : w \text{ even} \right\},$$

reproduced bit-for-bit.

5.3 Cross-check against the HSF Julia oracle

The pre-existing HSF code builds the sparse unitary and extracts Krylov sectors by connected components on the symmetrised adjacency (valid for unitary rules). Its `_H_SectorData` files use $V = H$, obc0. Matching its rule numbering (HSF 0–15; 6=150, 1=201, 2=198, 4=156, 8=108), **35/35 (rule, N) sector-size multisets match exactly** for $N = 8\text{--}14$, and rule 150 additionally at $N = 15, 16, 17$. Representative largest-sector ($D_{\max}$) sequences (obc0):

Wolfram (HSF)	$D_{\max}(N)$	character
150 (6)	126, 210, 462, 924, 1716, $\dots$, 43758	central binomials
201 (1)	55, 89, 144, 233, 377, 610, $\dots$	Fibonacci (φ)
108 (8)	21, 34, 55, 89, 144, $\dots$	shifted Fibonacci
156 (4) = 198 (2)	12, 16, 20, 27, 36, 48, 64, $\dots$	base $\approx 4^{1/5}$

That 156 and 198 share sector data confirms the reflection reduction (they are a reflection pair). The Fibonacci $D_{\max}$ of 201 is the Floquet-PXP golden-ratio scaling.

6 Symmetry note: rules 201 vs. 108 (broken inversion)

Rules 201 = (V, I, I, I) and 108 = (I, I, I, V) are $0 \leftrightarrow 1$ spin-flip (complement) partners. Classically the complement is a symmetry, but here the flip also conjugates the gate by X , and $XHX \neq \pm H$. We find, exactly:

- **pbc**: 201 and 108 have *identical* sector sizes for all N tested (47, 76, 123, 199, 322, 521, $\dots$). Because $|\langle y | XHX | x \rangle| = |\langle y | H | x \rangle|$, the bit-flip $X^{\otimes N}$ (a basis permutation) preserves the support graph, hence the sector partition.
- **obc0**: 201 and 108 *differ* ($D_{\max}$ Fibonacci vs. shifted Fibonacci, as in HSF rules 1 vs. 8). The fixed vacuum-0 padding is mapped to vacuum-1 by the bit-flip, so the boundary breaks the symmetry.

The full dynamics (spectra, entanglement) is spin-flip-broken *everywhere* because $XHX \neq \pm H$ changes amplitude signs. Consequently the pipeline never reduces V-containing rules by spin-flip. Reflection sector-size invariance is likewise used only for unitary rules; under the even-first brick-wall convention it does not extend to dissipative rules (reflection swaps the two layers).

7 Reproducibility

The engine is deterministic (no RNG in Tier 1a). Results are append-only JSON lines `results/{rule}_{bc}.js` keyed by `(engine_version, rule, N, bc)`, with an audit trail in `checkpoints/manifest.jsonl`; a killed run resumes without recomputation. Regression items are encoded as `pytest` tests under `code/qca_fragmentation/tests/`. This report rests on engine_version 1a.1.